Mentored Research Grant (MRG)

Project Title

Experiment-Driven Design of Multi-Material 3D-Printed Tensegrity Structures for Energy Absorption

Principal Investigator(s)

PI Name: Jeffrey R. Hill
PI Department: Mechanical Engineering
Co-PI: Sterling G. Baird, Department of Mechanical Engineering

Abstract

We propose a two-year experiment-driven mentored research program in which 2 undergraduate students design, fabricate, and test multi-material 3D-printed tensegrity-inspired structures for energy absorption. Using PLA for rigid struts and TPU for flexible tension elements on a single FDM platform, students will rapidly print and test many candidate geometries, generating high-quality physical data. A Bayesian optimization loop operating directly on experimental measurements will guide the search toward optimal energy-absorbing designs. The project is structured around layered mentoring—weekly one-on-one faculty meetings, lab group sessions, and graduate co-mentoring—that progressively moves students from guided tasks to independent research ownership. Students will gain hands-on experience in CAD design, multi-material 3D printing, mechanical testing, and data-driven optimization. Expected outcomes include undergraduate conference presentations (UCUR, ASME IDETC), a peer-reviewed journal submission, and development of transferable technical skills. This project provides a rich mentored learning environment at the intersection of structural mechanics, advanced manufacturing, and machine learning.

Number of Students to be Mentored

Number of Undergraduate Students: 2
Number of Graduate Students: 1 (co-mentor, in support of creating the mentoring environment)

Budget

Item	Year 1	Year 2	Total
Undergraduate wages	$9,000	$9,000	$18,000
Graduate wages	$1,250	$1,250	$2,500
Supplies	$2,250	$750	$3,000
Travel	$0	$1,500	$1,500
Total	**$12,500**	**$12,500**	**$25,000**

Relationship to External Funding

The PIs do not currently hold external funding for tensegrity or architected materials research. This MRG provides the primary support for establishing a mentored undergraduate research program in this area. If successful, the preliminary data may inform a future NSF proposal; however, the MRG is self-contained and its outcomes do not depend on external funding.

1 Research Motivation and Overview

We propose a two-year **experiment-driven** mentored research program in which 2 undergraduate students design, fabricate, and test **multi-material 3D-printed tensegrity-inspired structures** for energy absorption. Tensegrity (tensional integrity) structures—assemblies of rigid compression members suspended within a continuous network of tension members—exhibit remarkable strength-to-weight ratios and tunable mechanical responses, making them attractive for protective equipment, packaging, and aerospace applications [Skelton and de Oliveira, 2009, Fraternali et al., 2015]. Recent work has shown that *tensegrity-inspired* architectures can be directly 3D-printed and retain desirable load-limiting behavior under impact [Pajunen et al., 2019]. Because multimaterial FDM TPU elements are not ideal cables, we target tensegrity-inspired architectures that reproduce key tensegrity-like mechanical responses while remaining manufacturable; notably, TPU's rate-dependent damping is an *asset* for energy absorption rather than merely a deviation from idealized behavior.

Multi-material 3D printing with PLA (rigid struts) and TPU (flexible tension elements) enables rapid fabrication of diverse tensegrity-inspired geometries on a single print platform [Ye et al., 2023, Khatri and Egan, 2024]. Rather than relying on costly simulation campaigns, we leverage this manufacturing agility to pursue a **high-throughput experimental approach**: students will print, test, and iterate on many candidate designs—we estimate 50–100+ unique specimens over two years—generating high-quality physical data directly. **Bayesian optimization** (BO) will guide the experimental search, selecting the most informative designs to print and test next based on prior results [Shahriari et al., 2016, Frazier, 2018]. This experiment-first strategy produces reliable, real-world performance data while keeping the project accessible to undergraduates at all levels—students can contribute meaningfully through hands-on printing and testing without requiring advanced modeling expertise.

A primary objective of this project is to provide a rich, interdisciplinary mentored research experience for undergraduates. Through weekly one-on-one mentoring, structured lab meetings, and a progressive scaffolding model that moves students from guided tasks to independent research ownership, each student will gain hands-on skills in CAD, multi-material 3D printing, mechanical testing, and data-driven optimization. Each student will be responsible for defining design variables, executing experimental or computational campaigns, analyzing data, presenting results, and drafting manuscript sections.

2 Background

Tensegrity structures consist of isolated compression members (struts) held in equilibrium by a continuous network of tension members [Skelton and de Oliveira, 2009, Sultan, 2009]. Their defining characteristic—no two rigid bars touch—yields lightweight assemblies with surprising load-bearing capacity and tunable nonlinear responses [Amendola et al., 2014, Zhang et al., 2015]. Pajunen et al. [Pajunen et al., 2019] demonstrated that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact. Multi-material rigid–soft printing [Ye et al., 2023, Khatri and Egan, 2024] has shown that PLA–TPU combinations printed on a single FDM platform achieve tunable energy absorption. Ye et al. demonstrated a wrapping-based strategy (rigid cores encapsulated by continuous soft skins) that prevents interface delamination and enables cyclic durability—a critical enabler for our fabrication approach. These manufacturing advances mean that undergraduates can design, print, and test many configurations in rapid succession, making 3D printing the natural engine of experimental exploration.

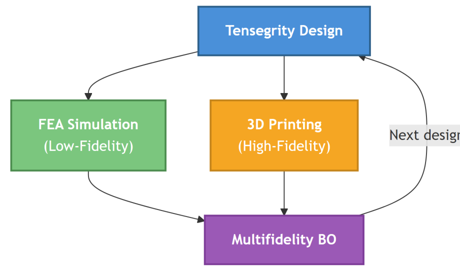

Figure 1: Research framework overview.

Bayesian optimization (BO) is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian process surrogate [Shahriari et al., 2016, Frazier, 2018]. BO has driven breakthroughs across domains—from tuning hyperparameters of AlphaGo [Silver et al., 2016] to optimizing sustainable concrete formulations [Ament et al., 2023] and discovering state-of-the-art organic laser emitters [Strieth-Kalthoff et al., 2024]—and has been applied to metamaterial design [Wang et al., 2022]. In our framework, BO operates *directly on experimental data*: each round of printing and testing provides observations that update a Gaussian process surrogate, which then recommends the next set of designs to fabricate. This closed-loop print–test–optimize cycle eliminates the need for simulation as a prerequisite and keeps the workflow centered on physical experimentation—an approach that is both scientifically rigorous and naturally suited to mentoring undergraduates at all experience levels.

3 Student Research Project 1: Design, Fabrication, and Testing

Student profile: One sophomore or junior undergraduate. No prior simulation or advanced modeling experience is required; the hands-on nature of this project makes it an ideal entry point for students early in their studies.

Scope and Deliverables

This student will lead the design-and-build cycle that forms the experimental backbone of the project. Working under weekly one-on-one mentoring from Prof. Hill, the student will begin by **parameterizing a family of tensegrity-inspired unit cells** in CAD (SolidWorks or Fusion 360), defining key variables—strut diameter and length (PLA), tension-element cross-section (TPU), connectivity topology, and unit-cell tiling—that span the design space. Following Ye et al. [2023], the designs use a core-wrapping strategy (rigid PLA struts encapsulated by continuous TPU skins) to ensure robust mechanical interlocking and prevent interface delamination. Because PLA–TPU interfaces are known to exhibit mixed-mode fracture behavior and potential void formation, print parameter validation and interface robustness will be explicitly evaluated during early testing.

The student will then **fabricate structures** using the department's multi-material FDM printer, systematically iterating on print parameters (layer height, infill, temperature) to achieve reliable geometries in both PLA and TPU. By printing many candidate designs quickly, the student builds an expanding library of tested specimens and develops practical expertise in additive manufacturing—a transferable skill valued in both graduate research and industry. The student works closely with Student 2, who tests each batch and returns performance data; this tight collaboration teaches both students the value of interdisciplinary teamwork and creates natural peer-mentoring opportunities as each develops complementary expertise. Throughout, Prof. Baird's master's student serves as a day-to-day co-mentor, guiding the student through equipment operation, troubleshooting print failures, and interpreting early results, so that learning happens at multiple cadences: weekly faculty meetings for strategy and daily peer support for execution.

Mentoring outcomes: The student will develop skills in CAD, multi-material additive manufacturing, and systematic experimental design. Structured mentoring scaffolds move the student from guided replication of existing designs to independent creation of novel geometries by mid-Year 1. The student will present results at **UCUR** or **ASME IDETC** and contribute as co-author on a journal manuscript.

4 Student Research Project 2: Mechanical Testing and Data-Driven Optimization

Student profile: One sophomore, junior, or senior undergraduate with coursework or interest in mechanics of materials or data analysis.

Scope and Deliverables

This student will lead physical testing and the Bayesian optimization campaign, working under weekly one-on-one mentoring from Prof. Baird. The work begins with **conducting experimental testing** on structures fabricated by Student 1: quasi-static compaction tests and drop-weight impact tests following the protocol of Pajunen et al. [2019]. Primary metrics include *peak transmitted force*, *specific energy absorption* (SEA), and *compaction efficiency*—tangible, operationally measurable outcomes well suited to undergraduate research.

With an initial data set in hand, the student will **implement a single-fidelity Bayesian optimization loop** (BoTorch/Ax) that operates directly on experimental measurements. After each BO iteration the surrogate recommends the next batch of designs; the student coordinates with Student 1 to print and test them, closing the loop. This closed-loop *design–print–test–optimize* cycle teaches the student data-driven decision-making in a concrete, physical context. If time and student interest permit, lightweight simulations may be added as an optional low-fidelity data source to accelerate convergence [Mo et al., 2023], but the campaign does not depend on simulation fidelity.

Prof. Baird's master's student provides day-to-day co-mentoring in Python, data analysis, and BO implementation, creating a layered mentoring structure. In Year 2, both students mentor newly recruited undergraduates, reinforcing their own understanding while sustaining the research pipeline.

Mentoring outcomes: The student will develop skills in experimental mechanics (test design, data acquisition), scientific computing (Python), and data-driven optimization. They will present results at **ASME IDETC** or **UCUR** and contribute as co-author on a journal manuscript.

5 Mentoring Environment

Each student in this program follows a structured path from guided exploration to independent research ownership, supported by layered mentoring at every stage. Because the project is centered on hands-on experimentation—designing, printing, and testing physical structures—students engage with tangible problems from day one, building confidence and competence through rapid, concrete feedback loops.

Recruitment

Students will be recruited from courses in mechanics of materials, machine design, kinematics, and compliant mechanisms, targeting sophomores through seniors. The experiment-driven nature of the project lowers the barrier to entry: students need not have prior simulation or programming experience, broadening the pool of potential mentees.

Mentoring Structure

Students meet individually with Profs. Hill and Baird each week (30 min) for technical guidance and professional development, and participate in weekly lab group meetings (1 hr) that include progress updates, student literature presentations, and skill-building workshops covering scientific writing, presentation skills, and research ethics. Prof. Baird's master's student serves as a graduate co-mentor, providing day-to-day guidance in 3D printing, testing procedures, coding, and data interpretation; this layered model ensures timely support while the graduate student develops their own mentoring skills. Students begin with structured tasks—replicating known tensegrity geometries and running standard test protocols—and progress to independent design decisions, culminating in ownership of their research project. Year 1 focuses on technical skill acquisition and guided execution; Year 2 emphasizes independent hypothesis generation, experimental design, and peer mentoring. In Year 2, experienced students mentor newly recruited sophomores, reinforcing their own knowledge and creating a sustainable pipeline of trained researchers.

Student Development Outcomes

The experiment-driven workflow structures mentoring around concrete milestones: a student's first successful multi-material print, first compression test, first data set fed into the BO loop. On the *technical* side, students gain experience in experimental mechanics (test design, data acquisition), additive manufacturing (CAD, multi-material 3D printing with wrapping strategies), and data-driven optimization (Python, Bayesian optimization). *Communication* skills develop through lab notebooks, group presentations, conference talks, and co-authoring a peer-reviewed manuscript. *Professional* development includes project management, data-driven decision-making (interpreting optimization recommendations and selecting next experiments), interdisciplinary collaboration, and preparation for graduate school or industry careers.

6 Expected Research Outcomes

The primary outcome of this MRG is the **training of 2 undergraduate researchers** with hands-on experience in multi-material 3D printing, mechanical testing, and data-driven optimization, prepared for graduate study or engineering careers. By completion, each undergraduate is expected to have participated in the design and testing of approximately 100+ tensegrity specimens, contributed to at least one **conference presentation** at UCUR and ASME IDETC and contributed as a co-author on a peer-reviewed manuscript submitted to the *ASME Journal of Mechanical Design* or *Smart Materials and Structures*. By centering the project on experimentation, students will each complete dozens of design–print–test cycles over two years, building deep practical competence through iterative, mentor-guided problem solving. The experimental data set—spanning a wide range of tensegrity-inspired geometries with measured energy-absorption performance—will be published on GitHub alongside optimization scripts for reproducibility. In Year 2, students will mentor incoming undergraduates, developing leadership skills and creating a self-sustaining research group.

7 Potential Impact of Work

This project advances architected material design for energy absorption with applications in **protective equipment** (helmets, body armor), **packaging** (shock-absorbing inserts), and **aerospace** (lightweight impact-resistant structures). By generating a large, systematically collected experimental data set of tensegrity-inspired structures, we provide the community with a resource for benchmarking and further optimization. The experiment-driven Bayesian optimization framework demonstrates that undergraduates can meaningfully drive a closed-loop experimental campaign, establishing a replicable mentoring model for data-driven research in other metamaterial classes. The layered mentoring structure—faculty, graduate co-mentor, and peer mentoring—can be adopted by other research groups seeking to involve undergraduates in hands-on engineering research. Open-source dissemination of data, code, and tested designs extends impact beyond BYU and enables other groups—including primarily undergraduate institutions—to build on this work.

8 Timeline

The project spans two academic years plus two summers (Table 1). **Year 1** focuses on experimental foundations: student recruitment and onboarding in Fall, followed by literature review, tensegrity parameterization in CAD, and initial multi-material print trials through Winter. The first Summer term is a critical period of full-time research with intensive mentoring—faculty increase meeting frequency to twice weekly, and the graduate co-mentor works alongside students daily. Students ramp up fabrication throughput, run systematic compression and drop-weight impact tests, and begin feeding results into the Bayesian optimization loop. **Year 2** shifts to optimization and dissemination: BO-guided experimental campaigns continue through Fall and Winter, with students printing and testing progressively refined designs. Students present findings at UCUR and ASME IDETC in Year 2, and a journal manuscript is prepared during Winter and Summer. Throughout, experienced students mentor newly recruited undergraduates, reinforcing their own skills and ensuring continuity of expertise.

9 Budget

Total requested: **$25,000** over 2 years. Per grant guidelines, no funds are allocated to equipment, tuition, or faculty wages.

Task	Y1 F	Y1 W	Y1 Su	Y2 F	Y2 W	Y2 Su
Student recruitment & onboarding	•					
Literature review	•	•				
Tensegrity parameterization & CAD	•	•				
Print trials & process validation	•	•	•			
Multi-material 3D printing campaigns		•	•	•	•	
Compression & impact testing		•	•	•	•	
Bayesian optimization loop			•	•	•	
Conference presentations (UCUR/IDETC)				•	•	
Journal manuscript preparation					•	•
Peer mentoring of new students				•	•	

Table 1: Project timeline across two years (F = Fall, W = Winter, Su = Summer). Summer terms support full-time research with intensive mentoring. Year 2 includes formal peer mentoring, where experienced students supervise newly recruited undergraduates.

Category	Description	Year 1	Year 2
Undergraduate wages	2 students, part-time + summer full-time	\$9,000	\$9,000
Graduate student wages	Partial RA for grad co-mentor	\$1,250	\$1,250
Supplies	Filament, load cells, DAQ, consumables	\$2,250	\$750
Travel	Conference registration & travel	\$0	\$1,500
Annual Total		**\$12,500**	**\$12,500**

Undergraduate wages (\$18,000): Supports 2 undergraduates working part-time (∼10 hrs/week) during fall and winter semesters and full-time (∼40 hrs/week) during summer terms for simulation, 3D printing, and testing. **Graduate student wages (\$2,500):** Prof. Baird's master's student serves as day-to-day co-mentor. **Supplies (\$3,000):** PLA/TPU filament (sufficient for 50–100+ specimens), load cells, DAQ module, fasteners, poster printing, and other consumables (Year 1 heavy to support initial equipment and print trials). **Travel (\$1,500):** Student presentations at ASME IDETC or UCUR (Year 2).

References

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Biographical Sketches

Jeffrey R. Hill — Principal Investigator

Professional Preparation

Ph.D.	Mechanical Engineering	Pennsylvania State University (2011)
M.S.	Mechanical Engineering	Brigham Young University (2005)
B.S.	Mechanical Engineering	Brigham Young University (2004)

Appointments and Positions

- Associate Professor, Dept. of Mechanical Engineering, Brigham Young University (2021–present)
- Principal Member of the Technical Staff, Sandia National Laboratories (2016–2021)
- Senior Member of the Technical Staff, Sandia National Laboratories (2011–2016)
- Visiting Research Investigator, University of Michigan (2008–2011)

Products Most Closely Related to This Proposal

1. Layer B, Denning H, Hill J. Control and locomotion of tensegrity robots through manipulation of the center of mass. *Robotica*. 2024. DOI: 10.1017/S0263574724001139
2. Denning H, Bullinger A, Hill J. Tuning Tensegrity: Simplified Construction and Cable Tension Interactions. *ASME SMASIS 2024*. DOI: 10.1115/SMASIS2024-140381
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4. Masmeijer T, Swain C, Hill J, Habtour E. Programming tension in 3D printed networks inspired by spiderwebs. *Materials & Design*. 2025. DOI: 10.1016/j.matdes.2025.115281
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Other Significant Products

1. Hill J, Wang K. Development of the En Masse Elimination Algorithm for Actuator Grouping Optimization. *J. Intelligent Material Systems and Structures*. 2011;22(11):1203–1212. DOI: 10.1177/1045389X11417651
2. Kim J, Hill J, Wang K. An asymptotic approach for the analysis of piezoelectric fiber composite beams. *Smart Materials and Structures*. 2011;20(2):025023.
3. Hill J, Wang K, Fang H, Quijano U. Actuator grouping optimization on flexible space reflectors. *SPIE Active and Passive Smart Structures*. 2011. DOI: 10.1117/12.880086

Synergistic Activities

- Mentoring undergraduate and graduate students in experimental mechanics, tensegrity structures, and smart materials research at BYU.

- Teaching courses in mechanics of materials, machine design, and kinematics in the Department of Mechanical Engineering.
- Collaboration with Sandia National Laboratories on advanced materials characterization and structural design.

Sterling G. Baird — Co-Principal Investigator

Professional Preparation

Ph.D.	Materials Science and Engineering	University of Utah (2023)
M.S.	Mechanical Engineering	Brigham Young University (2021)
B.S.	Applied Physics	Brigham Young University (2018)

Appointments and Positions

- Assistant Professor, Dept. of Mechanical Engineering, Brigham Young University (2025–present)
- Principal Investigator and Staff Scientist, Acceleration Consortium, University of Toronto (2025)
- Director of Training and Programs, Acceleration Consortium, University of Toronto (2023–2025)

Products Most Closely Related to This Proposal

1. Baird SG, Falkowski AR, Sparks TD. Honegumi: An Interface for Accelerating the Adoption of Bayesian Optimization in the Experimental Sciences. 2025. DOI: 10.26434/chemrxiv-2024-c70v7-v2
2. Baird SG, Liu M, Sparks TD. High-Dimensional Bayesian Optimization of 23 Hyperparameters over 100 Iterations for an Attention-Based Network to Predict Materials Property. *Computational Materials Science*. 2022;211:111505. DOI: 10.1016/j.commatsci.2022.111505
3. Baird SG, Hall JR, Sparks TD. Compactness Matters: Improving Bayesian Optimization Efficiency of Materials Formulations through Invariant Search Spaces. *Computational Materials Science*. 2023;224:112134. DOI: 10.1016/j.commatsci.2023.112134
4. Tom G, Schmid SP, Baird SG, et al. Self-Driving Laboratories for Chemistry and Materials Science. *Chemical Reviews*. 2024;124(16):9633–9732. DOI: 10.1021/acs.chemrev.4c00055
5. Baird SG, Sparks TD. Building a “Hello World” for Self-Driving Labs: The Closed-loop Spectroscopy Lab Light-mixing Demo. *STAR Protocols*. 2023;4(2):102329. DOI: 10.1016/j.xpro.2023.102329

Other Significant Products

1. Lo S, Baird SG, et al. Review of Low-Cost Self-Driving Laboratories in Chemistry and Materials Science: The “Frugal Twin” Concept. *Digital Discovery*. 2024;3(5):842–868. DOI: 10.1039/D3DD00223C

2. Baird SG, et al. Bayesian Optimization Hackathon for Chemistry and Materials. 2025. DOI: 10.26434/chemrxiv-2025-dzh5z
3. Seegmiller CC, Baird SG, Sayeed HM, Sparks TD. Discovering Chemically Novel, High-Temperature Superconductors. *Computational Materials Science*. 2023.

Synergistic Activities

- Development of open-source tools for Bayesian optimization, including Honegumi (interactive BO script generator) and contributions to the Ax Platform.
- Mentoring of 8 undergraduate research assistants at BYU and 12+ staff and students at the Acceleration Consortium.
- Organization of the Acceleration Consortium Bayesian Optimization Hackathon (120 participants, 62 organizations, 14 countries).
- Guest Editor for *npj Computational Materials*; journal reviewer for *Nature Communications*, *Digital Discovery*, *JOSS*.